

Fauzan Mohammed

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SUMMARY

Experienced Data Scientist with a focus on **ML/NLP technologies**, adept at building scalable AI systems. Proven track record in developing NLP pipelines and implementing evaluation frameworks, ready to contribute to Penn State University's mission of translating nutrition literature into actionable insights.

EDUCATION

Rensselaer Polytechnic Institute

Masters in IT/CS

Troy, NY

Aug 2023 – Dec 2024

Kwame Nkrumah University of Science and Technology

Bachelor of Science in Mechanical Engineering

Kumasi, Ghana

Sep 2016 – June 2020

EXPERIENCE

Data Scientist – NLP/LLM Systems + Responsible AI

Walmart Global Tech

Aug 2024 – Present

Sunnyvale, CA

- Contributed to the design and deployment of a production multi-agent LLM system powering Walmart's conversational shopping platform, orchestrating intelligent search, guided discovery, and personalized recommendations for 20M+ monthly active users.
- Designed prompt-tuned LLM-as-judge evaluation frameworks for search quality and product relevance assessment across raw and reranked results, driving measurable improvements in ranking precision across product categories.
- Built production guardrail infrastructure using prompt-tuned LLM judges on GCP and Element Gateway, implementing real-time safety enforcement and hallucination detection to ensure responsible AI compliance for 10M+ users.
- Designed and maintained prompt templates, multi-step chains, and persona configurations across agent workflows, implementing version-controlled prompt libraries for consistency and reuse across the conversational platform.
- Developed an AI agent-based Event Title Generator leveraging LLM function calling to trigger internal APIs, autonomously infer user intent, map to structured event categories, and generate personalized titles within the shopping experience.
- Architected multi-turn conversation management logic including memory handling, contextual continuity, and dynamic persona assignment across domain-specific agent roles.
- Developed unsupervised NLP pipeline using DBSCAN clustering for topic modeling and intent extraction, expanding chatbot query coverage from 15% to 74%.

Data Scientist, Intern

Walmart Global Tech

May 2024 – Aug 2024

Sunnyvale, CA

- Engineered Transformer-based fallback detection system using semantic clustering to classify 100K+ daily fallback queries into 27 intent clusters, improving diagnostic accuracy by 6%.
- Optimized LLM router architecture through prompt engineering and query decomposition, improving task-selection accuracy by 4% and reducing latency by 18%.
- Built automated evaluation dashboards using SQL and **Python** for real-time monitoring of precision, recall, and prompt drift, reducing experimentation cycles by 30%.

Junior Machine Learning Engineer

Omdena Thrive

Dec 2022 – Apr 2023

Remote

- Engineered BERT-based NLP models with attention mechanisms achieving 85% classification accuracy on 5K+ mental health journal entries; deployed production analytics dashboard on GCP using Flask and Plotly for 200+ clinical users.

PROJECTS

AI-Powered Financial Rule Automation System

Dec 2024

- Built AI rule-generation engine integrating multiple Transformer models (MiniLM, RoBERTa, DeBERTa) with QuickBooks API, implementing multi-task fusion architecture to auto-categorize transactions at 92% precision and reduce manual rule creation by 70%.

LEADERSHIP

Co-founder and Director

Fama Foundation

Aug 2019 – Present

Accra, Ghana

- Led impactful programs across 10 Ghanaian communities, enrolling 325 teenage girls in computer programming and literacy education.

TECHNICAL SKILLS

Languages: Python, SQL, Java

ML/DL: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, spaCy, NLTK

LLM & AI: LangChain, LlamaIndex, FAISS, OpenAI/Anthropic/Gemini APIs

Cloud & Data: GCP (Vertex AI, BigQuery, Cloud Run), Docker, Kubernetes, Pandas, NumPy, Spark

MLOps: MLflow, Weights & Biases, Flask, FastAPI, CI/CD, Git, DVC

Specialties: Multi-Agent Systems, LLM Fine-Tuning (SFT, RLHF, DPO), Prompt Engineering, RAG, Model Evaluation, Responsible AI, Function Calling & Tool Use, Conversation Flow Design